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Inventor(s)	Peyrot; Jacques et al.

Osteosynthesis device

Abstract

In one embodiment, the present invention includes an intramedullary osteosynthesis or arthrodesis implant including a central body, a first pair of legs extending from the central body to a pair of leg tips, and a second pair of legs extending from the central body, in a direction opposite the first pair of legs, to a pair of leg tips such that the central body, first pair of legs and second pair of legs are positioned alongside a longitudinal axis.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
321389	12/1884	Schirmer	N/A	N/A
1095054	12/1913	Wiesenfeld	N/A	N/A
1517334	12/1923	Young	N/A	N/A
1893864	12/1932	Kocher	N/A	N/A
2128005	12/1937	Lombard	N/A	N/A
2208848	12/1939	Jorgensen	N/A	N/A
2531911	12/1949	Johnson	N/A	N/A
2580821	12/1951	Nicola	N/A	N/A
2984248	12/1960	Sidelman	N/A	N/A
3338689	12/1966	Hetzel et al.	N/A	N/A
3462765	12/1968	Swanson	N/A	N/A
3466669	12/1968	Flatt	N/A	N/A
3593342	12/1970	Niebauer et al.	N/A	N/A
3646654	12/1971	Cervenka et al.	N/A	N/A
3681786	12/1971	Lynch	N/A	N/A
3739403	12/1972	Nicolle	N/A	N/A
3805302	12/1973	Mathys	N/A	N/A
3824631	12/1973	Burstein et al.	N/A	N/A
3875594	12/1974	Swanson	N/A	N/A
D243716	12/1976	Treace et al.	N/A	N/A

4091806	12/1977	Aginsky et al.	N/A	N/A
4158893	12/1978	Swanson	N/A	N/A
4204284	12/1979	Koeneman	N/A	N/A
4276660	12/1980	Laure	N/A	N/A
4364382	12/1981	Mennen	N/A	N/A
4367562	12/1982	Gauthier et al.	N/A	N/A
4485816	12/1983	Krumme	N/A	N/A
D277509	12/1984	Lawrence et al.	N/A	N/A
D277784	12/1984	Sgarlato et al.	N/A	N/A
4522200	12/1984	Stednitz	N/A	N/A
D284099	12/1985	Laporta et al.	N/A	N/A
4634382	12/1986	Kusano et al.	N/A	N/A
D291731	12/1986	Aikins	N/A	N/A
4759768	12/1987	Hermann et al.	N/A	N/A
4871367	12/1988	Christensen et al.	N/A	N/A
4905679	12/1989	Morgan	N/A	N/A
4955916	12/1989	Carignan et al.	N/A	N/A
4969909	12/1989	Barouk	N/A	N/A
5011497	12/1990	Persson et al.	N/A	N/A
5047059	12/1990	Saffar	N/A	N/A
5062851	12/1990	Branemark	N/A	N/A
5074865	12/1990	Fahmy	N/A	N/A
5092896	12/1991	Meuli et al.	N/A	N/A
5108443	12/1991	Branemark	N/A	N/A
5133761	12/1991	Krouskop	N/A	N/A
5179915	12/1992	Cohen et al.	N/A	N/A
5190546	12/1992	Jervis	N/A	N/A
5207712	12/1992	Cohen	N/A	N/A
5326364	12/1993	Clift, Jr. et al.	N/A	N/A
5360450	12/1993	Giannini	N/A	N/A
5405400	12/1994	Linscheid et al.	N/A	N/A
5405401	12/1994	Lippincott, III et al.	N/A	N/A
5417692	12/1994	Goble et al.	N/A	N/A
5425776	12/1994	Cohen	N/A	N/A
5425777	12/1994	Sarkisian et al.	N/A	N/A
5454814	12/1994	Comte	N/A	N/A
5464427	12/1994	Curtis et al.	N/A	N/A
5474557	12/1994	Mai	N/A	N/A
D366114	12/1995	Ohata	N/A	N/A
5480447	12/1995	Skiba	N/A	N/A
5484443	12/1995	Pascarella et al.	N/A	N/A
D369412	12/1995	Morgan	N/A	N/A
5507822	12/1995	Bouchon et al.	N/A	N/A
5522903	12/1995	Sokolow et al.	N/A	N/A
5554157	12/1995	Errico et al.	N/A	N/A
5578036	12/1995	Stone et al.	N/A	N/A
5634925	12/1996	Urbanski	N/A	N/A
5674297	12/1996	Lane et al.	N/A	N/A
5690631	12/1996	Duncan et al.	N/A	N/A

5702472	12/1996	Huebner	N/A	N/A
D388877	12/1997	Morgan	N/A	N/A
5725585	12/1997	Zobel	N/A	N/A
5779707	12/1997	Bertholet et al.	N/A	N/A
5782927	12/1997	Klawitter et al.	N/A	N/A
5824095	12/1997	Di Maio, Jr. et al.	N/A	N/A
5876434	12/1998	Flomenblit et al.	N/A	N/A
5881443	12/1998	Roberts et al.	N/A	N/A
5882444	12/1998	Flomenblit et al.	N/A	N/A
5919193	12/1998	Slavitt	N/A	N/A
5951288	12/1998	Sawa	N/A	N/A
5958159	12/1998	Prandi	N/A	N/A
5984970	12/1998	Bramlet	N/A	N/A
5984971	12/1998	Faccioli et al.	N/A	N/A
6011497	12/1999	Tsang et al.	N/A	N/A
6017366	12/1999	Berman	N/A	N/A
6093188	12/1999	Murray	N/A	N/A
6123709	12/1999	Jones	N/A	N/A
6146387	12/1999	Trott et al.	N/A	N/A
6162234	12/1999	Freedland et al.	N/A	N/A
6187008	12/2000	Hamman	N/A	N/A
6193757	12/2000	Foley et al.	N/A	N/A
6197037	12/2000	Hair	N/A	N/A
6200330	12/2000	Benderev et al.	N/A	N/A
6248109	12/2000	Stoffella	N/A	N/A
6261289	12/2000	Levy	N/A	N/A
6319284	12/2000	Rushdy et al.	N/A	N/A
6325805	12/2000	Ogilvie et al.	N/A	N/A
6342076	12/2001	Lundborg	N/A	N/A
6348052	12/2001	Sammarco	N/A	N/A
6352560	12/2001	Poeschmann et al.	N/A	N/A
6383223	12/2001	Baehler et al.	N/A	N/A
6386877	12/2001	Sutter	N/A	N/A
6423097	12/2001	Rauscher	N/A	N/A
6428634	12/2001	Besselink et al.	N/A	N/A
6454808	12/2001	Masada	N/A	N/A
6475242	12/2001	Bramlet	N/A	N/A
6554833	12/2002	Levy et al.	N/A	N/A
6689169	12/2003	Harris	N/A	N/A
6692499	12/2003	Tormala et al.	N/A	N/A
6699247	12/2003	Zucherman et al.	N/A	N/A
6699292	12/2003	Ogilvie et al.	N/A	N/A
6706045	12/2003	Lin et al.	N/A	N/A
6736818	12/2003	Perren et al.	N/A	N/A
6773437	12/2003	Ogilvie et al.	N/A	N/A
6811568	12/2003	Minamikawa	N/A	N/A
6827741	12/2003	Reeder	N/A	N/A
6869449	12/2004	Ball et al.	N/A	N/A
6896177	12/2004	Carter	N/A	N/A
6981974	12/2005	Berger	N/A	N/A

7025789	12/2005	Chow et al.	N/A	N/A
7037342	12/2005	Nilsson et al.	N/A	N/A
7041106	12/2005	Carver et al.	N/A	N/A
7044953	12/2005	Capanni	N/A	N/A
7052498	12/2005	Levy et al.	N/A	N/A
7182787	12/2006	Hassler et al.	N/A	N/A
7240677	12/2006	Fox	N/A	N/A
7291175	12/2006	Gordon	N/A	N/A
7588603	12/2008	Leonard	N/A	N/A
7600956	12/2008	McDuff et al.	N/A	N/A
7655042	12/2009	Foley et al.	N/A	N/A
7780737	12/2009	Bonnard et al.	N/A	N/A
7794483	12/2009	Capanni	N/A	N/A
7837738	12/2009	Reigstad et al.	N/A	N/A
7842091	12/2009	Johnstone et al.	N/A	N/A
7909880	12/2010	Grant	N/A	N/A
7918879	12/2010	Yeung et al.	N/A	N/A
7922765	12/2010	Reiley	N/A	N/A
7955388	12/2010	Jensen et al.	N/A	N/A
7976580	12/2010	Berger	N/A	N/A
8048173	12/2010	Ochoa	N/A	N/A
8100983	12/2011	Schulte	N/A	N/A
8162942	12/2011	Coati et al.	N/A	N/A
8202305	12/2011	Reiley	N/A	N/A
8262712	12/2011	Coilard-Lavirotte et al.	N/A	N/A
8308779	12/2011	Reiley	N/A	N/A
8394097	12/2012	Peyrot et al.	N/A	N/A
8414583	12/2012	Prandi et al.	N/A	N/A
8475456	12/2012	Augoyard et al.	N/A	N/A
8529611	12/2012	Champagne et al.	N/A	N/A
8597337	12/2012	Champagne	N/A	N/A
8608785	12/2012	Reed et al.	N/A	N/A
8685024	12/2013	Roman	N/A	N/A
8715325	12/2013	Weiner et al.	N/A	N/A
8840623	12/2013	Reiley	N/A	N/A
8864804	12/2013	Champagne et al.	N/A	N/A
9011504	12/2014	Reed	N/A	N/A
9125698	12/2014	Miller	N/A	N/A
9282977	12/2015	Penzimer et al.	N/A	N/A
9283007	12/2015	Augoyard et al.	N/A	N/A
9554914	12/2016	Taylor et al.	N/A	N/A
9724140	12/2016	McCormick	N/A	N/A
9757168	12/2016	Seavey et al.	N/A	N/A
9775630	12/2016	Leavitt et al.	N/A	N/A
10111690	12/2017	Anderson et al.	N/A	N/A
2001/0025199	12/2000	Rauscher	N/A	N/A
2001/0049529	12/2000	Cachia et al.	N/A	N/A
2002/0019636	12/2001	Ogilvie et al.	N/A	N/A
2002/0055785	12/2001	Harris	N/A	N/A

2002/0065561	12/2001	Ogilvie et al.	N/A	N/A
2002/0068939	12/2001	Levy et al.	N/A	N/A
2002/0082705	12/2001	Bouman et al.	N/A	N/A
2002/0099395	12/2001	Acampora et al.	N/A	N/A
2002/0133156	12/2001	Cole	N/A	N/A
2002/0169066	12/2001	Cassidy et al.	N/A	N/A
2002/0189622	12/2001	Cauthen et al.	N/A	N/A
2003/0040805	12/2002	Minamikawa	N/A	N/A
2003/0069645	12/2002	Ball et al.	N/A	N/A
2003/0120277	12/2002	Berger	N/A	N/A
2003/0130660	12/2002	Levy et al.	N/A	N/A
2004/0002759	12/2003	Ferree	N/A	N/A
2004/0093081	12/2003	Nilsson et al.	N/A	N/A
2004/0102853	12/2003	Boumann et al.	N/A	N/A
2004/0138756	12/2003	Reeder	N/A	N/A
2004/0172031	12/2003	Rubecamp et al.	N/A	N/A
2004/0220574	12/2003	Pelo et al.	N/A	N/A
2004/0220678	12/2003	Chow et al.	N/A	N/A
2004/0230193	12/2003	Cheung et al.	N/A	N/A
2005/0119757	12/2004	Hassler et al.	N/A	N/A
2005/0124990	12/2004	Teague et al.	N/A	N/A
2005/0251265	12/2004	Calandruccio et al.	N/A	N/A
2005/0261768	12/2004	Trieu	623/17.11	A61B 17/7065
2005/0283159	12/2004	Amara	N/A	N/A
2006/0015181	12/2005	Elberg	N/A	N/A
2006/0052725	12/2005	Santilli	N/A	N/A
2006/0052878	12/2005	Schmieding	N/A	N/A
2006/0074492	12/2005	Frey	N/A	N/A
2006/0084998	12/2005	Evy et al.	N/A	N/A
2006/0085075	12/2005	McLeer	N/A	N/A
2006/0247787	12/2005	Rydell et al.	N/A	N/A
2007/0038303	12/2006	Myerson et al.	N/A	N/A
2007/0123993	12/2006	Hassler et al.	N/A	N/A
2007/0142920	12/2006	Niemi	N/A	N/A
2007/0156241	12/2006	Reiley et al.	N/A	N/A
2007/0166122	12/2006	McDuff	411/387.1	F16B 13/061
2007/0185584	12/2006	Kaufmann et al.	N/A	N/A
2007/0198088	12/2006	Biedermann et al.	N/A	N/A
2007/0213831	12/2006	de Cubber	N/A	N/A
2007/0233110	12/2006	Muhanna et al.	N/A	N/A
2007/0239158	12/2006	Trieu et al.	N/A	N/A
2008/0039949	12/2007	Meesenburg et al.	N/A	N/A
2008/0132894	12/2007	Coilard-Lavirotte et al.	N/A	N/A
2008/0154385	12/2007	Trail et al.	N/A	N/A
2008/0177262	12/2007	Augoyard et al.	N/A	N/A
2008/0177291	12/2007	Jensen et al.	N/A	N/A
2008/0195219	12/2007	Wiley et al.	N/A	N/A
2008/0221697	12/2007	Graser	N/A	N/A

2008/0221698	12/2007	Berger	N/A	N/A
2008/0234763	12/2007	Patterson et al.	N/A	N/A
2008/0269908	12/2007	Warburton	N/A	N/A
2009/0005821	12/2008	Chirico et al.	N/A	N/A
2009/0012564	12/2008	Chirico et al.	N/A	N/A
2009/0138096	12/2008	Myerson et al.	N/A	N/A
2009/0254189	12/2008	Scheker	N/A	N/A
2009/0254190	12/2008	Gannoe et al.	N/A	N/A
2010/0010637	12/2009	Pequignot	N/A	N/A
2010/0016905	12/2009	Greenhalgh et al.	N/A	N/A
2010/0016982	12/2009	Solomons	N/A	N/A
2010/0057214	12/2009	Graham et al.	N/A	N/A
2010/0121390	12/2009	Kleinman	N/A	N/A
2010/0131014	12/2009	Peyrot	N/A	N/A
2010/0131072	12/2009	Schulte	N/A	N/A
2010/0161068	12/2009	Lindner et al.	N/A	N/A
2010/0185295	12/2009	Emmanuel	N/A	N/A
2010/0228301	12/2009	Greenhalgh et al.	N/A	N/A
2010/0249942	12/2009	Goswami et al.	N/A	N/A
2010/0256770	12/2009	Hakansson et al.	N/A	N/A
2010/0262254	12/2009	Lawrence et al.	N/A	N/A
2011/0004317	12/2010	Hacking et al.	N/A	N/A
2011/0093084	12/2010	Morton	N/A	N/A
2011/0093085	12/2010	Morton	N/A	N/A
2011/0118739	12/2010	Tyber et al.	N/A	N/A
2011/0144644	12/2010	Prandi et al.	N/A	N/A
2011/0301652	12/2010	Reed et al.	N/A	N/A
2011/0301653	12/2010	Reed et al.	N/A	N/A
2012/0029579	12/2011	Bottlang et al.	N/A	N/A
2012/0065692	12/2011	Champagne et al.	N/A	N/A
2012/0083791	12/2011	Cheney et al.	N/A	N/A
2012/0197311	12/2011	Kirschman	N/A	N/A
2013/0053975	12/2012	Reed et al.	N/A	N/A
2013/0060295	12/2012	Reed et al.	N/A	N/A
2013/0066435	12/2012	Averous et al.	N/A	N/A
2013/0131822	12/2012	Lewis et al.	N/A	N/A
2013/0150965	12/2012	Taylor et al.	N/A	N/A
2014/0058462	12/2013	Reed et al.	N/A	N/A
2014/0142715	12/2013	McCormick	N/A	N/A
2014/0180428	12/2013	McCormick	N/A	N/A
2014/0188239	12/2013	Cummings	N/A	N/A
2015/0112342	12/2014	Penzimer et al.	N/A	N/A
2015/0223849	12/2014	McCormick et al.	N/A	N/A
2016/0058484	12/2015	McCombs- Stearnes et al.	N/A	N/A
2016/0338751	12/2015	Kellar et al.	N/A	N/A
2017/0252084	12/2016	Anderson et al.	N/A	N/A
2017/0333081	12/2016	Cordier et al.	N/A	N/A
2018/0021145	12/2017	Seavey et al.	N/A	N/A
2018/0161170	12/2017	Petranto	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2551021	12/2004	CA	N/A
2243699	12/2005	CA	N/A
2836654	12/2013	CA	N/A
2837497	12/2013	CA	N/A
0042808	12/1980	EP	N/A
0340159	12/1988	EP	N/A
0420794	12/1990	EP	N/A
0454645	12/1990	EP	N/A
1300122	12/2002	EP	N/A
1356794	12/2002	EP	N/A
1582159	12/2004	EP	N/A
1923012	12/2007	EP	N/A
2663838	12/1991	FR	N/A
2725126	12/1995	FR	N/A
2783702	12/1999	FR	N/A
2787313	12/1999	FR	N/A
2794019	12/1999	FR	N/A
2801189	12/2000	FR	N/A
2846545	12/2003	FR	N/A
2856269	12/2003	FR	N/A
2884406	12/2005	FR	N/A
2119655	12/1982	GB	N/A
2430625	12/2006	GB	N/A
S60145133	12/1984	JP	N/A
03001854	12/1990	JP	N/A
H7303662	12/1994	JP	N/A
2004535249	12/2003	JP	N/A
3648687	12/2004	JP	N/A
2007530194	12/2006	JP	N/A
2008188411	12/2007	JP	N/A
2008537696	12/2007	JP	N/A
4695511	12/2010	JP	N/A
5631597	12/2013	JP	N/A
20070004513	12/2006	KR	N/A
20070022256	12/2006	KR	N/A
101004561	12/2010	KR	N/A
101235983	12/2012	KR	N/A
9116014	12/1990	WO	N/A
9625129	12/1995	WO	N/A
9641596	12/1995	WO	N/A
9726846	12/1996	WO	N/A
9733537	12/1996	WO	N/A
0117445	12/2000	WO	N/A
03084416	12/2002	WO	N/A
2005020830	12/2004	WO	N/A
2005020831	12/2004	WO	N/A
2005063149	12/2004	WO	N/A

2005104961	12/2004	WO	N/A
2006109004	12/2005	WO	N/A
2007135322	12/2006	WO	N/A
2008057404	12/2007	WO	N/A
2008112308	12/2007	WO	N/A
2009103085	12/2008	WO	N/A
2011130229	12/2010	WO	N/A

OTHER PUBLICATIONS

International Search Report for PCT/FR2008/050453 dated Nov. 4, 2008, 4 pages. cited by applicant

International Search Report, PCT/FR2006/050345, dated Aug. 30, 2006, 3 pages. cited by applicant

Cross section, <<https://byjus.com/maths/cross-section/>> (last visited Jan. 26, 2022). 4 pgs. [Exhibit No. 1009 to Petition for Inter Partes Review of U.S. Pat. No. 9,168,074]. cited by applicant

Declaration of Michael Sherman (Jan. 28, 2022). 119 pgs. [Exhibit No. 1002 to Petition for Inter Partes Review of U.S. Pat. No. 9,168,074]. cited by applicant

Petition for Inter Partes Review of U.S. Pat. No. 9,168,074 , *OsteoMed LLC v. Stryker European Operations Holdings LLC*. (Jan. 28, 2022). 98 pgs. cited by applicant

Pietrzak WS, et al., “A bioabsorbable fixation implant for use in proximal interphalangeal joint (hammer toe) arthrodesis: Biomechanical testing in a synthetic bone substrate”. *J Foot Ankle Surg.* Sep.-Oct. 2006. 45(5):288-94. doi: 10.1053/j.jfas.2006.05.004. PMID: 16949524. 7 pgs. [Exhibit No. 1007 to Petition for Inter Partes Review of U.S. Pat. No. 9,168,074]. cited by applicant

The American Heritage College Dictionary, Fourth Edition, Houghton Mifflin Company (Apr. 2007). 3 pgs. [Exhibit No. 1008 to Petition for Inter Partes Review of U.S. Pat. No. 9,168,074]. cited by applicant

Patent Owner's Preliminary Response and Exhibits List, IPR2022-00486 of U.S. Pat. No. 9,168,074 , *OsteoMed LLC v. Stryker European Operations Holdings LLC*. (Filed May 16, 2022), 77 pages. [Including Appendices at Exhibits 2003 and 2004]. cited by applicant

Collins English Dictionary Excerpt (Jun. 2007), 6 pages. [Exhibit No. 2001 to Patent Owner's Preliminary Response, IPR2022-00486 of U.S. Pat. No. 9,168,074 filed May 16, 2022]. cited by applicant

Excerpt from Tool.com—File and Rasp Tools, (Copyright 2022), 6 pages. [Exhibit No. 2002 to Patent Owner's Preliminary Response, IPR2022-00486 of U.S. Pat. No. 9,168,074 filed May 16, 2022]. cited by applicant

Jung, H. J. et al., JJ., Decision Denying Institution of Inter Partes Review, IPR2022-00486 of U.S. Pat. No. 9,168,074 , *OsteoMed LLC v. Stryker European Operations Holdings LLC*. (Aug. 12, 2022). 42 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 15/802,794, filed Nov. 3, 2017, which is a continuation of U.S. application Ser. No. 13/686,074, filed Nov. 27, 2012, now U.S. Pat. No. 9,839,453, which is a continuation of U.S. application Ser. No. 12/531,577, filed Feb. 1, 2010, now U.S. Pat. No. 8,394,097, which is a national phase entry under 35 U.S.C. § 371 of International Application No.

FIELD OF THE INVENTION

(1) The invention relates to the technical field of orthopaedic implants, in particular for arthrodeses and osteosyntheses.

BACKGROUND OF THE INVENTION

(2) It may be recalled that an osteosynthesis implant must serve to hold in place two (or more) parts of the same bone fractured or cut by a surgical operation (osteotomy), for the time necessary for the consolidation of this bone (typically 3 months).

(3) It may be recalled that an arthrodesis is the blocking of a joint by surgery to fuse two bones into a single one, using an osteosynthesis device.

(4) It may be recalled that the purpose of any osteosynthesis and particularly in the case of an arthrodesis is to obtain very good primary and also secondary stability, in order to obtain the consolidation in the best possible conditions, that is, in a position selected by the surgeon, by minimizing the problems of post-operative pain and edemas, by shortening the consolidation time as much as possible.

(5) To obtain this result, the stability of the osteosynthesis associated with the implant is critical. Furthermore, the implant must also provide and maintain a slight compression on the portions of bone to be fused together, thereby facilitating this consolidation.

(6) Various technical solutions have been proposed to carry out an arthrodesis, in particular at the tips (foot, hand, wrist, ankle, etc.).

(7) Mention can be made, for example of basic staples which do not provide a proper fixation during consolidation, and shape memory staples which serve to place the two bone portions to be consolidated under compression, thereby corresponding to the intended purpose.

(8) However, to obtain satisfactory stability, it is necessary to place two, or even three staples, in different planes. This significantly increases their size, thereby limiting the applications, in particular on small bones (for example in the fingers or toes).

(9) It is also common to use extramedullary or extra-osseous plates and screws, which also require a relatively large space and cannot be used on the terminal phalanges of the fingers (distal interphalangeal arthrodesis for example). Moreover, the medium-term stability of these systems is not always guaranteed (loosening of the mounting).

(10) Certain types of screw can be used in intramedullary cases, but in this case, the approach path requires a pulpar approach, which may generate serious complications (sepsis, etc.) and discomfort for the patient.

(11) Use can also be made of pins which have a smaller size. However, the stability obtained is not satisfactory (problems of migration) and it is generally necessary to remove them after consolidation. With such devices, moreover, the patient cannot immerse the finger or toe treated, because the pin generally projects outside the skin.

(12) Intramedullary osteosynthesis systems are available for long bones (tibia, femur, humerus, etc.). For example, lockable centromedullary nails are known. Apart from the fact that the locking technique is difficult, it cannot be miniaturized for extremity surgery (hand and foot).

(13) Shape memory intramedullary devices are also available for solving part of the problems with respect to the arthrodesis or the osteosynthesis of the small fragments: for example, the solutions described in French patent 2 846 545 or French patent 2 884 406 (US 2008/0177262).

(14) French Patent 2 846 545 describes an H-shaped device which opens in the body into an X, thanks to the use of a shape memory set around 37° C., each leg being implanted in a calibrated hole.

(15) In practice, such a system does not allow proper introduction into the bone. This is because the preparation of 2 parallel holes in a phalange is extremely difficult due to the limited size and, above

all, the parallel legs tend to open naturally during introduction and thereby exert an effect of distraction of the two fragments rather than compression.

(16) Furthermore, the use of shape memory is very limiting due to the demands it makes on surgeons, in particular of temperature management: the implant must be fitted into the bone when cold before it warms and opens. This requires placing the implant in a support, storing it cold, and using all possible speed for implantation.

(17) Finally, since the legs are straight, their shape memory tends to create a local support at their tips, which does not ensure satisfactory behavior and can damage the bone.

(18) US 2008/0177262 teaches a system for easier introduction whether by the shape (eye) or by a support or a clamp that keeps the legs of the implant closed during introduction.

(19) Nevertheless, these systems do not operate very dependably, because they do not define the optimal criteria allowing proper introduction into the bone and good anchorage: the anchoring zones always tend to open too early, thereby blocking introduction.

OBJECT OF THE INVENTION

(20) It is the object of the invention to remedy all these drawbacks simply, reliably, effectively and efficiently.

(21) The problem that the invention proposes to solve is to define the success criteria for an intramedullary implant, easy to place and effective for generating primary and secondary stability of the osteosynthesis or arthrodesis focus thanks to its stiffness and its compression component.

SUMMARY OF THE INVENTION

(22) The inventive implant is characterized in that it comprises two bone anchoring zones on either side of a stiff stability zone, withstanding shear forces, these two anchoring zones having a possibility of high deformation at their base (in particular by elasticity) and a design such that they can adopt a closed position (in particular thanks to a suitable clamp closed at their base) for easy introduction into a calibrated centromedullary hole (prepared with an appropriate instrument), and in that owing to this particular configuration, they offer in the bone site the possibility of obtaining the final impaction without a distraction effect on the bone and sufficient expansion to ensure proper fixation in the bone.

(23) The anchoring zones can be deformed at their base by elasticity, superelasticity or shape memory and typically consist of branches or legs, optionally connected (olive or rugby ball shape). In the open shape, these branches have an outward positive angle at their base and are curved inwardly toward their tips, whereas when the shape is closed, the angle of the base is reversed, that is negative or turned inwardly, thereby obtaining the width at the tips (impaction side) that is smaller than their base, in order to avoid impaction in the bone and blocking of the penetration of the implant.

(24) The invention has a particularly advantageous application, which cannot however be considered as limiting, for the preparation of arthrodeses in the phalanges of the fingers and toes, especially for the proximal and distal interphalangeal joints in the hand and foot.

(25) The device is implantable via the dorsal path (or optionally lateral or palmar/plantar), but without pulpar approach, thereby minimizing the risks of infection and improving the patient's comfort.

(26) To take account of the anatomical features, the anchoring zones are connected to the median zone serving for strength (in particular shear) at the osteosynthesis focus by more or less long connecting zones, and the central zone may have a bend to adapt to the characteristics of the desired arthrodesis.

(27) The material constituting the implant of the invention must allow a certain minimal opening of the anchoring zones once the implant is in the body. It may therefore be made from any sufficiently elastic implantable material such as stainless steel, titanium, a bioresorbable material such as polylactide acid PLLA.

(28) Preferably, the implant of the invention is prepared from a shape memory material used for its

property of superelasticity (or elasticity associated with the transformation of the austenite martensite phase under stress) which has the widest known elasticity range (up to 8% in elastic elongation equivalent in traction for an implant of Nitinol, a nickel-titanium alloy comprising about 55.5 to 56% by weight of nickel, the remainder being titanium).

(29) It is also possible to use a material having a thermal shape memory around 37° C.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention is described in greater detail below in conjunction with the appended drawings in which:

(2) FIG. 1 shows an exemplary implant of the invention in its full open position in 3 dimensions,

(3) FIG. 2 shows the same example in its closed introduction position in its main plane,

(4) FIG. 3 shows an exemplary implant of the invention in its full open position, in its main plane,

(5) FIG. 4 shows an example of a bent implant according to the invention in its full open position,

(6) FIGS. 5 (5a to 5f) shows the implantation sequence: closed, half introduced still stressed, fully introduced on one side (free in the bone), and the same sequence on the other side.

SPECIFIC DESCRIPTION

(7) The implant is in the form of 2 anchoring zones A1 and A2 connected by a central zone C (FIG. 1) and optionally intermediate connecting zones such that in the closed position, the shape is substantially inscribed in a very elongated rectangle (FIG. 2), and in the open shape, corresponds to a wider X-shape due to the spreading of the anchoring zones A1 and A2 (FIG. 3).

(8) The anchoring zones A1 and A2 each have two legs P1 and P2 having lengths L1 and L2 (FIG. 3).

(9) The cross-section of the implant is adapted to the implantation sites, but preferably flat in order to have good mechanical strength and reduced size (typically the thickness e is about 1 to 2 mm) (FIG. 1).

(10) FIG. 2 shows the closed position with the various widths of the implant: L_{ab} is the width of the central zone C, L_{1ab} and L_{2ab} are the widths at the base of the anchoring zones A1 and A2. These 3 widths may be equal or slightly different to adapt to the bone site. Typically, the widths are about 2 to 5 times the thickness (or 2 to 10 mm). These dimensions are adapted to the various dimensions of the hand and foot but are not limiting because they depend on the bone site of the operated patient.

(11) The anchoring zones A1 and A2 are suitable for separation by elastic effect or by shape memory effect at their base, so that the maximum width in the open position at the tips $La1$ and $La2$ (FIG. 3) is at least equal to the width of the base of the same anchoring zone in the closed position plus 50% minimum, or plus a minimum of 1.5 mm. This means that $La1 > L_{1ab} + 50\%$ or $La1 > L_{1ab} + 1.5$ mm and that $La2 > L_{2ab} + 50\%$ or $La2 > L_{2ab} + 1.5$ mm. This opening criterion is necessary to have sufficient fixation in the bone.

(12) As shown in FIG. 3, the legs P1 and P2 are substantially straight at their base (on about $\frac{1}{3}$ to half of their length) and are then rounded inwardly at their tips (on about $\frac{1}{3}$ to half of their length). In the open position, the straight inner portions of the legs P1 and P2 make positive outward angles $a1$ and $a2$ with the longitudinal implant axis A (FIG. 3), whereas in the closed position, these angles become inward negative angles $b1$ and $b2$ (FIG. 2). The upper or outer portions (toward the tip) of the legs P1 and P2 virtually undergoes no particular deformation between the two open and closed shapes.

(13) This particular geometric arrangement ensures that in the closed position, the legs virtually touch at the tips (FIG. 2), and that the widths at the tips in the closed position $La1f$ and $La2f$ are lower than the widths at the respective bases L_{1ab} and L_{2ab} , thereby allowing easy introduction

without distraction of the distal bone fragment and also obtaining the opened/closed movement by a local deformation at the base of the legs, that is by leaving the distal zone free for introducing this zone into the bone.

(14) In order to obtain both easy introduction and sufficient opening movement, the angles α_1 and α_2 are preferably between +5 degrees and +25 degrees and the angles β_1 and β_2 between 0 degrees and -15 degrees.

(15) Preferably, the width of the tips of the anchoring zones in the closed position La_{1f} and La_{2f} are lower than the widths of the bases of the zones L_{1ab} and L_{2ab} , minus 20%: $La_{1f} < L_{1ab} - 20\%$ and $La_{2f} < L_{2ab} - 20\%$.

(16) The legs or anchoring zones are thus “articulated” at their base, and can therefore be secured in the closed position on a support or even better a clamp, positioned at an appropriate location defined in particular in the case of an elastic material (for a shape memory material, this is not absolutely necessary since the shape does not change as long as the activation temperature is not reached), this clamp not covering more than half of the length of the legs, thereby allowing introduction of at least half of the implant into its recess.

(17) The inside tangents at the tip of the legs P1 and P2 in the open position make angles β_1 and β_2 with the longitudinal axis A of the implant close to 0 degrees, in order to have a good bone contact area along the whole lengths of the legs in the open position and to prevent the bone from being touched by the tips alone (FIG. 3).

(18) In the implant site, at body temperature, the implant can still be in the closed position, or parallel or with semiopen legs so that the force exerted by opening of the legs is transmitted to the bone and ensures proper fixation.

(19) This “olive” arrangement of the legs, associated with an “articulation” of the base and associated with a minimal introduction of half of their length allows completion of the insertion, once the clamp has been removed.

(20) In order to guarantee satisfactory operation, the elasticity or memory of the piece must allow a transition from the closed shape (typically width 2 to 4 mm according to the size of the site) to an open shape with a significant movement (+1.5 to +3 mm approximately).

(21) Similarly, the force of expansion of the legs (or swelling of the olive) must be significant: typically 1 to 3 kg for an arthrodesis of the tips (force measured at 37° C. in the blocked introduction position), without being excessive: it is important for the legs to avoid opening completely and for the bone to resist so as to have a real holding force.

(22) The legs P1 and P2 or fins may have a rough surface or even better notches D (FIG. 3) on their outer surfaces intended to be press into the spongy bone and form a good anchorage. The typical depth of these notches H1 and H2 is about 0.5 mm. The opening of the legs must be at least 1.5 times this depth in order to ensure good engagement of the notches in the bone or 1.5 mm.

(23) The legs P1 and P2 may also have a surface covered with an osseointegration coating such as hydroxyapatite (HAP) intended to facilitate the anchorage.

(24) To facilitate introduction into the bone, the tips of the legs P1 and P2 are bevelled with an inward angle to the longitudinal axis A of the implant W1 and W2 (FIG. 3). This angle is typically between -20 degrees and -40 degrees.

(25) By tests on fresh cadavers and experience, an optimal level of the force was determined with a minimum allowing anchorage of the notches in the spongy bone and a maximum force to be certain to avoid damaging the implantation site.

(26) After tests and experience, an ideal zone was found with a maximum 20% of the elastic limit of the bone measured in a blocked closed shape at 37° C., which, considering the dimensions of the implant, gives rise to maximum values of about 3 kg, and the need for a rapid lowering as soon as the anchorage is obtained, or a force divided by 2 in the semi-open position (a force of 0.5 to 1.5 kg allows good holding). In fact, if the opening force is higher than about 3 kg, introduction into the bone becomes much more difficult, or even impossible above 4 kg. Finally, in order to guarantee a

damage-free site, it is necessary for the force to become negligible for a virtually complete opening. These values are indicative and depend on the arthrodesis site and the bone quality.

(27) In one version of the invention, the notches D1 and D2 on the outside of the legs P1 and P2 allow the positioning of a clamp and introduction at the base of the legs P1 and P2 (FIG. 3). These notches are symmetrical by pairs of legs and their spacing d is the same on the legs P1 and on the legs P2. A first notch D1 has a first sidewall (10) and a second sidewall (12), and a second notch D1 has a third sidewall (16) and a fourth sidewall (14) as shown in FIG. 3.

(28) The central zone C must have a minimum length L_c equal to the length d between the notches D1 and D2 so that even in case of movement of the implant during final impaction, this zone C remains in the arthrodesis focus and performs its resistance function.

(29) In one version of the invention, an orifice Or is provided in this central zone for positioning a holding pin to prevent migration of the implant at the time of final impaction.

(30) As shown in FIG. 4, this central zone may be bent at an angle Ag defined between the 2 main planes formed by the legs P1 on the one hand and P2 on the other hand to adapt to the surgical requirements for adjusting the position of the arthrodesis. In most cases, the angle Ag is fixed between 0° (typically flat position for an index) and 30° (typically for a little finger).

(31) As an example, an operating technique of implantation of the inventive device for the case of an elastic or superelastic implant is described as follows as shown in FIG. 5:

DETAILED DESCRIPTION OF THE FEATURES OF THE INVENTION AND EXEMPLARY EMBODIMENTS

(32) Approach by dorsal path Resection of cartilages and osteophytes Centromedullary perforation using an appropriate instrument to make a calibrated rectangular hole having a width of substantially $L1ab$ or $L2ab$ and thickness of substantially e (by a suitable rasp) Closure of the clamp side P1 (FIG. 5a) Implant introduction side P1 to minimum half (FIG. 5b) Clamp removal Complete introduction side P1 (FIG. 5c) Closure of the clamp side P2 (FIG. 5d) Placement of the bone side P2 on the implant side P2 to about half (FIG. 5e) Removal of the clamp Manual impaction of the bone side P2 on the bone P1 (FIG. 5f)

(33) In a particular embodiment, intended for a distal interphalangeal arthrodesis (hand), the implant is prepared from a superelastic Nitinol alloy (nickel-titanium in the weight proportion 55.8% nickel and 44.2% titanium).

(34) The cross-section of the central zone C is $Lab \times e = 2.8 \times 1.2$ mm and the legs are asymmetrical to adapt better to the shapes of the bone, minimize the implanted metal section and allow sufficient expansion for good anchorage. The length of the legs is $L2 = 6.5$ mm distal side P2 and $L1 = 9$ mm proximal side P1. The length of the central zone C is 3 mm, allowing a slight offset during closure, without affecting the shear strength. To adapt to the surgeon's choice, this central zone may be bent (typically flat or 15° or 25°).

(35) In the closed position, the width of the proximal base $L1ab$ is 3.8 mm and of the distal base $L2ab$ is 3.0 mm. The opening of the legs P1 and P2 is 2.5 mm or 2.2 mm, that is $La1$ is 6.3 mm and $La2$ is 5.2 mm. In the open position the angle at the base of the legs is $a1 = 10^\circ$ and $a2 = 22^\circ$. The straight portion is about 45% of the total length. The curvature of the distal tip of the legs is calculated so that the angle of the tangent at the tip is $\beta1 = -5^\circ$ and $\beta2 = -3^\circ$. In the closed position, the angle at the base of the legs is $b1 = -4^\circ$, $b2 = -2^\circ$. And the width at tip is $La1f = 2.5$ mm and $La2f = 2.1$ mm.

(36) In one embodiment of the invention, the 0.5 mm deep notches are distributed on the legs (1 notch at approximately 0.8 mm spacings).

(37) The angle of incidence of the tip of the legs (including notches) is $w1 = 33^\circ$ and $w2 = 24^\circ$, allowing easy introduction without the distraction effect between the two bone pieces to be osteosynthesized.

(38) The rounded design of the anchoring zones serves to obtain a maximized contact area over the entire length in the open shape, with an impaction effect in the spongy bone, and hence a spongy

packing effect.

(39) In another example, more appropriate for arthrodesis of the thumb, the dimensions are rather the following:

(40) Closed widths: $L1ab=6.5$ mm, $L2ab=5$ mm, with an opening of 3 to 4 mm approximately to obtain: $La1=11$ mm and $La2=8$ mm and $L1=13$ mm and $L2=9$ mm.

Claims

1. An intramedullary osteosynthesis or arthrodesis implant comprising: a monolithic body extending along a longitudinal axis, the body including a plurality of notches on an outward-facing surface of the body, at least a first notch of the plurality of notches is spaced from a second notch of the plurality of notches in a first direction along the axis, the first and second notches extending outwardly in the first direction from the axis, and at least the first notch having a taper along the axis in a different direction than a taper of the second notch along the axis, the first notch being wider adjacent to the monolithic body and narrower at an opposite end thereof to define the taper of the first notch, the second notch being wider adjacent to the monolithic body and narrower at an opposite end thereof to define the taper of the second notch, wherein the first notch is defined by first and second sidewalls extending from the body, the first sidewall being longer than the second sidewall, the second notch being defined by third and fourth sidewalls, the third sidewall being longer than the fourth sidewall, the second sidewall facing the fourth sidewall such that second sidewall is parallel to the fourth sidewall, wherein the monolithic body includes a pair of legs, the first and second notches on a first leg of the pair of legs, the pair of legs being movable about a base point.
2. The implant of claim 1, wherein the pair of legs meet at the base point.
3. The implant of claim 1, wherein an inside surface of the pair of legs is free of notches.
4. The implant of claim 1, wherein a second leg of the pair of legs has a third notch and a fourth notch of the plurality of notches, the third notch and the fourth notch extending in a second direction opposite the first direction, the third notch having a taper along the axis in a different direction than a taper of the fourth notch.
5. An intramedullary osteosynthesis or arthrodesis implant comprising: a monolithic body extending along a longitudinal axis, the body including a plurality of notches on an outward-facing surface of the body, at least a first notch of the plurality of notches is spaced from a second notch of the plurality of notches in a direction along the axis, the first and second notches extends outwardly in a first direction from the axis, the first notch being wider adjacent to the monolithic body and narrower at a tip at an opposite end thereof, the second notch being wider adjacent to the monolithic body and narrower at a tip at an opposite end thereof, the first notch being defined by first and second sidewalls extending from the body, the first sidewall being longer than the second sidewall, the second notch being defined by third and fourth sidewalls, the third sidewall being longer than the fourth sidewall, the second sidewall facing the fourth sidewall such that second sidewall is parallel to the fourth sidewall, wherein, in an open position, the tip of the first notch extending outwardly a first distance from the axis and the tip of the second notch extending outwardly a second distance from the axis, the first distance is greater than the second distance, wherein the first and second notches are movable about a base point from the open position to a closed position.
6. The implant of claim 5, wherein the monolithic body includes a pair of legs, the first and second notches on a first leg of the pair of legs, the pair of legs being movable about a base point from the open position to a closed position.
7. The implant of claim 6, wherein the pair of legs meet at the base point.
8. The implant of claim 6, wherein an inside surface of the pair of legs is free of notches.
9. The implant of claim 6, wherein a second leg of the pair of legs has a third notch and a fourth

notch of the plurality of notches, the third notch and the fourth notch extending in a second direction opposite the first direction, wherein, in the open position, the tip of the third notch extends outwardly a third distance from the axis and a tip of the fourth notch extends outwardly a fourth distance from the axis, the third distance being greater than the fourth distance.

10. The implant of claim 9, wherein the first distance is the same as the third distance and the second distance is the same as the fourth distance.

11. The implant of claim 9, wherein the first leg has a fifth notch of the plurality of notches and the second leg has a sixth notch of the plurality of notches, the fifth notch extending in the first direction and the sixth notch extending in the second direction, the fifth notch extends outwardly a fifth distance from the axis and the sixth notch extends outwardly a sixth distance from the axis, the fifth distance being less than the first distance and the second distance, the sixth distance being less than the third distance and the fourth distance.

12. An intramedullary osteosynthesis or arthrodesis implant comprising: a monolithic body extending along a longitudinal axis, the body including a plurality of notches on an outward-facing surface of the body, at least a first notch of the plurality of notches is spaced from a second notch of the plurality of notches in a direction along the axis, the first and second notches extending outwardly in the same direction from the axis, the first notch having a first taper toward a tip of the first notch and the second notch having a second taper toward a tip of the second notch, the first taper being longer than the second taper, the first notch being wider adjacent to the monolithic body and narrower at an opposite end thereof to define the first taper, the second notch being wider adjacent to the monolithic body and narrower at an opposite end thereof to define the second taper, wherein the first notch is defined by first and second sidewalls extending from the body, the first sidewall being longer than the second sidewall, the second notch being defined by third and fourth sidewalls, the third sidewall being longer than the fourth sidewall, the second sidewall facing the fourth sidewall such that second sidewall is parallel to the fourth sidewall, wherein the monolithic body includes a pair of legs, the first and second notches on a first leg of the pair of legs, the pair of legs being movable about a base point.

13. The implant of claim 12, wherein the first notch is at a tip of the monolithic body.

14. The implant of claim 12, wherein the pair of legs meet at the base point.

15. The implant of claim 12, wherein an inside surface of the pair of legs is free of notches.

16. The implant of claim 12, wherein a second leg of the pair of legs having a third notch and a fourth notch of the plurality of notches, the third notch and the fourth notch extending in a second direction opposite the first direction, wherein the third notch has a third taper and the fourth notch has a fourth taper, the third taper being longer than the fourth taper.

17. The implant of claim 12, wherein an outside surface of the monolithic body defines a space free of notches between the first notch and the second notch, the space having a first distance.

18. The implant of claim 17, wherein the second notch defines a second distance along the outside surface of the monolithic body, the first distance being greater than the second distance.
